

Problem 10.29

You are given that

$$d^{(2)} = 0.04.$$

Calculate δ .

Solution

The nominal annual discount rate $d^{(2)} = 0.04$ is convertible semi-annually, so the discount rate per half-year is

$$\frac{d^{(2)}}{2} = \frac{0.04}{2} = 0.02.$$

The accumulation factor for each half-year is therefore

$$\left(1 - \frac{d^{(2)}}{2}\right)^{-1}.$$

Since there are two half-year periods in one year, the annual accumulation factor is

$$\left(1 - \frac{d^{(2)}}{2}\right)^{-2}.$$

The equivalent constant force of interest δ satisfies

$$e^{\delta} = \left(1 - \frac{0.04}{2}\right)^{-2}.$$

Taking natural logarithms gives

$$\begin{aligned}\delta &= \ln \left[\left(1 - \frac{0.04}{2}\right)^{-2} \right] \\ &= -2 \ln(0.98) \\ &\approx 0.040405414635.\end{aligned}$$

Thus, the equivalent force of interest is

$$\boxed{\delta \approx 4.0405\%}.$$